

F-6. Establishment of Community Tree Nurseries				
Forest / Shrub Land Conservation	Medium	Medium		Term
Measure Type	Cost	Preferences by Community	Hazards	Implementation Time of Measures

Descriptions									
Measures	A group of individuals coming together to collect local tree seeds, plant them, and nurture them until they become young trees can be referred to as establishing a community tree nursery. After that, the trees can be planted in a number of places to form hedgerows, forests, orchards, or individual trees. Many local organizations, such as churches, schools, self-help groups, or private farmers, can own and run community tree nurseries. They can raise species that the local population is interested in, give seedlings at a reduced cost, and create prospects for money generation.								
	The establishment of community tree nurseries is a crucial phase in addressing the socio-economic and environmental issues specific to Bangladesh's CHT. Such nurseries need a customized strategy to succeed because of the mountainous and rough terrain, seasonal rainfall, and inadequate infrastructure. By choosing suitable species, increasing local capacity, and guaranteeing community participation, these nurseries can make significant improvements to sustainable land use practices and offer a reliable source of income for surrounding communities.								
Location	R-114	R-99	R-65	K-98	K-76	K-71	B-173	B-77	B-76
	■	■	■	■	■	■	■	■	■
Applicable Sites	- Flat to Gently Sloping Land (0-5%) – Ideal for nursery bed preparation and seedling care. - Near Water Sources – Ensures consistent irrigation and seedling survival. - Degraded Land for Rehabilitation – Supports reforestation and watershed restoration projects.								
Slope	Less than 3%		3% to 15%		15% to 30%		30% to 60%		Above 60%
	√		√						
Design options and performance	- Centralized Community Nursery – Managed by a local group, producing diverse tree species for community needs. - Small-Scale Decentralized Nurseries – Individual farmers grow trees at a household level for income generation. - Agroforestry-Based Nurseries – Integrates fruit, timber, and medicinal plants to enhance biodiversity and local livelihoods. - Seedling Distribution Hubs – Nurseries serve as seedling banks, ensuring availability for reforestation and conservation projects.								

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	Shade-Net Nurseries – Protects seedlings from intense sun and heavy rainfall, improving survival rates.			
Feasibility criteria	- To avoid soil erosion, the site should be situated in a moderately sloping, somewhat stable location with easy access to natural water sources for irrigation. In addition to being protected from severe weather, the selected location should have ideal soil, climate, and moisture conditions that promote the growth of healthy seedlings. - The species selected for the nursery should have ecological value, be native to the area, and be able to tolerate the local climate. Soil-stabilizing species and commercially valuable species, such as fruit trees, bamboo, and teak, should be prioritized. To achieve both ecological and socioeconomic objectives, a variety of species (such as timber, fuelwood, fruits, and medicinal plants) should be taken into account. - A long-term management strategy should be included of the project to guarantee the nursery's viability. This entails preserving seedling quality, modernizing species selection, and making sure funding is available for ongoing operations. For strategy adjustments and ongoing success, it is crucial to regularly monitor and assess nursery performance, community involvement, and impact on the environment.			
Operation and maintenance	In order to preserve the sustainability of the community tree nursery and the health and growth of the seedlings, it engages in a number of crucial activities. To safeguard the seedlings, regularly water and fertilize them, remove weeds, and use pest and disease control methods like biological or organic treatments. Frequent assessment and monitoring of nursery outputs, survival rates, and seedling health will help modify tactics and guarantee positive results.			
Cost	Medium			
Benefits				
	Type	Low	Medium	High
	Canopy Cover Increase			
	Carbon Sequestration			
	Crop Production Increase			
	Crop Failure Risk Reduction			
	Ensuring Food Security			
	Fruit Production Increase			
	Forestry Goods Production Increase			
	Livestock Production Increase			
	Fisheries Production Increase			
	Alternative Livelihood Generation			
	Increase of Income			
	Soil Erosion Control			
	Improved Soil Health			
	Soil Fertility Increase			
	Sustainable Water Supply			
	Water Quality Improvement			
	Water Purification			

	River Bank Erosion Control			
	Surface Runoff Reduction			
	Groundwater Recharge			
	Pest and Disease Management			
	Reduction of Pesticides and Fertilizer Use			
	Ecosystem and Biodiversity Increase			
	Better Communications			
	Better Education			
	Better WASH facilities			
Implementation Challenges				
	Type	Low	Medium	High
	Land unavailable/high land area required			
	Supply chain disconnection			
	Land right related problems			
	High human investment capital need			
	Insufficient infrastructures			
	Less demand of agroforestry product			
	Environmentally unsustainable			
	Socially unacceptable			
	Less beneficial			
Environmental performance	- Diverse native species planting from community nurseries maintain wildlife habitats and foster healthy ecosystems. These species are essential to preserving the natural equilibrium and enhancing the general well-being of forests. - Community tree nurseries are an essential component of efforts to slow down climate change since trees absorb carbon dioxide from the atmosphere. Trees absorb carbon as they grow, lowering the region's overall carbon footprint and stabilizing the climate globally. - Trees can contribute to climate regulation by reducing heat and encouraging moisture evaporation. The local population as well as the nearby plants and animals gain from this microclimatic influence, particularly in areas with harsh weather or unpredictable rainfall patterns. - By cultivating species that can be harvested for fuelwood, lumber, or other non-timber forest products, tree nurseries support sustainable forest management. By easing the strain on already-existing woods, this preserves ecological integrity and guarantees that local resources will be accessible to future generations.			
Remarks	This measure will aid landscape restoration in CHT of Bangladesh along with creating alternative livelihood opportunities.			